

Nonlinear Optimization: Challenges in ML

Exam-Pattern Practice Problems – Fully Worked, Step by Step

ZC416 Mathematical Foundations for ML/DS • Lecture 10 (Topic #10)

All questions use **fresh functions and data** in the style typically asked in quizzes, mid-semester and comprehensive exams. Every problem is numerical: compute, derive, or trace.

#	Skill tested	Difficulty	Marks
1	Min-max scaling vs. standardization (+ unseen point)	Easy	5
2	Update-step imbalance across parameters	Easy	4
3	Gradients before vs. after standardization	Moderate	8
4	Differential curvature: trace, stability, condition number	Moderate	8
5	Least-squares gradient: derive + one GD step	Moderate	6
6	Basins of attraction + random restarts	Difficult	6
7	Hessian before/after scaling: step-size gain	Difficult	8

Problem 1 (5 marks) **EASY** **COMPUTATION**

A feature column contains $\{30, 50, 70, 90, 110\}$.

- Apply min-max normalization $x' = \frac{x - \min}{\max - \min}$ to all five values. [1.5]
- Standardize the column (mean-center, divide by the population standard deviation). [2]
- A new observation $x = 130$ arrives at test time. Transform it under *both* schemes (using the training statistics) and comment on the results. [1.5]

Step-by-step solution

Step 1 (min-max). Here $\min = 30$, $\max = 110$, $\text{range} = 80$:

$$30 \mapsto 0, \quad 50 \mapsto \frac{20}{80} = 0.25, \quad 70 \mapsto 0.5, \quad 90 \mapsto 0.75, \quad 110 \mapsto 1.$$

Step 2 (standardize). Mean:

$$\mu = \frac{30 + 50 + 70 + 90 + 110}{5} = 70, \quad \text{deviations} = (-40, -20, 0, 20, 40).$$

Variance and standard deviation:

$$\sigma^2 = \frac{1600 + 400 + 0 + 400 + 1600}{5} = 800, \quad \sigma = \sqrt{800} \approx 28.28.$$

Standardized values:

$$z = \frac{x - 70}{28.28} : \quad -1.414, \quad -0.707, \quad 0, \quad 0.707, \quad 1.414.$$

(Check: the z -values sum to 0 and their squares sum to 5 $\Rightarrow$ unit variance. $\checkmark$)

Step 3 (unseen point $x = 130$). Min-max:

$$x' = \frac{130 - 30}{80} = 1.25 \text{ - outside } [0, 1].$$

The $[0, 1]$ guarantee holds only for training data; a downstream model assuming that range now extrapolates. Standardization:

$$z = \frac{130 - 70}{28.28} = 2.12 \quad \text{– unusual but well-defined (z-scores are unbounded by design).}$$

Answer. (a) 0, 0.25, 0.5, 0.75, 1; (b) $\mu = 70$, $\sigma \approx 28.28$, $z = (-1.414, -0.707, 0, 0.707, 1.414)$; (c) min-max gives 1.25 (out of range), z-score gives 2.12 (fine).

Intuition & exam tips

The part-(c) twist – transforming an *unseen* point with *training* statistics – is where marks are separated. Min-max promises $[0, 1]$ only in-sample and is fragile to outliers/new extremes; standardization degrades gracefully. Always use the training μ, σ ; never recompute statistics with the test point.

Problem 2 (4 marks) EASY COMPUTATION

At the current iterate, a loss has partial derivatives $\frac{\partial J}{\partial w_1} = -640$ and $\frac{\partial J}{\partial w_2} = -1.6$, and a single learning rate $\gamma = 10^{-3}$ is used for both parameters.

- (a) Compute the update to each parameter in this step. [1]
- (b) Assuming gradient components of this magnitude persist, roughly how many iterations does w_2 need to move by 1 unit? [1]
- (c) If instead γ is chosen so that the w_1 -update is a modest 0.01 per step, how many iterations does w_2 then need for a 1-unit move? [1]
- (d) Name the lecture technique that removes this dilemma and state (in one line) what it equalizes. [1]

Step-by-step solution

Step 1 (updates). Using $\Delta w_j = -\gamma \partial J / \partial w_j$:

$$\Delta w_1 = -10^{-3}(-640) = 0.64, \quad \Delta w_2 = -10^{-3}(-1.6) = 0.0016.$$

A 400:1 imbalance in movement per step.

Step 2 (the slow coordinate).

$$N \approx \frac{1}{0.0016} = 625 \text{ iterations}$$

for one unit of progress in w_2 – during which w_1 would have moved $625 \times 0.64 = 400$ units (long since overshoot and oscillating).

Step 3 (protecting w_1 instead). Require $\gamma \cdot 640 = 0.01$:

$$\gamma = 1.5625 \times 10^{-5} \quad \Rightarrow \quad \Delta w_2 = 1.5625 \times 10^{-5} \times 1.6 = 2.5 \times 10^{-5},$$

$$N \approx \frac{1}{2.5 \times 10^{-5}} = 40\,000 \text{ iterations.}$$

Step 4 (the fix). **Feature normalization / standardization** (mean-center and divide by the standard deviation, or min-max scale): it equalizes the loss's *sensitivity* (gradient scale / curvature) across parameters, so one shared learning rate suits all coordinates.

Answer. (a) 0.64 vs 0.0016; (b) ≈ 625 ; (c) $\approx 40,000$; (d) standardize features $\Rightarrow$ comparable per-parameter sensitivities.

Intuition & exam tips

The “one γ , two very different gradients” dilemma in pure arithmetic: protect the fast coordinate and the slow one crawls (40,000 steps); serve the slow one and the fast one oscillates. Whenever an exam gives you two wildly different partials, this cost-benefit computation is the expected answer skeleton.

Problem 3 (8 marks) MODERATE COMPUTATION

House price y (lakh) is modeled as $\hat{y} = w_1x_1 + w_2x_2$ from area x_1 (sq. ft) and bedroom count x_2 :

x_1	x_2	y
1000	2	50
1500	3	75
2000	3	90

with $J(w) = \sum_i (w_1x_{i1} + w_2x_{i2} - y_i)^2$.

- (a) Compute $\partial J/\partial w_1$ and $\partial J/\partial w_2$ at $w = (0, 0)$ and their ratio. [3]
- (b) Standardize both columns (population σ) and recompute the two partials at $w = (0, 0)$. [4]
- (c) Quantify the improvement and its consequence for gradient descent. [1]

Step-by-step solution

Step 1 (raw gradients). At $w = (0, 0)$ all predictions are 0, so $\frac{\partial J}{\partial w_j} = -2 \sum_i y_i x_{ij}$:

$$\frac{\partial J}{\partial w_1} = -2(1000(50) + 1500(75) + 2000(90)) = -2(342\,500) = -685\,000,$$

$$\frac{\partial J}{\partial w_2} = -2(2(50) + 3(75) + 3(90)) = -2(595) = -1190.$$

Ratio:

$$\frac{685000}{1190} \approx 576$$

– the loss is $\sim 576\times$ more sensitive to w_1 , purely because square-feet are numerically huge.

Step 2 (standardize column x_1). Mean and deviations:

$$\mu_1 = \frac{1000 + 1500 + 2000}{3} = 1500, \quad \text{deviations} = (-500, 0, 500).$$

$$\sigma_1 = \sqrt{\frac{250000 + 0 + 250000}{3}} \approx 408.25 \quad \Rightarrow \quad z_1 = (-1.2247, 0, 1.2247).$$

Step 3 (standardize column x_2).

$$\mu_2 = \frac{2 + 3 + 3}{3} = \frac{8}{3} \approx 2.667, \quad \text{deviations} = (-0.667, 0.333, 0.333).$$

$$\sigma_2 = \sqrt{\frac{0.4444 + 0.1111 + 0.1111}{3}} \approx 0.4714 \quad \Rightarrow \quad z_2 = (-1.4142, 0.7071, 0.7071).$$

Step 4 (recompute both partials). At $w = (0, 0)$:

$$\frac{\partial J}{\partial w_1} = -2((-1.2247)(50) + 0(75) + (1.2247)(90)) = -2(48.99) = -97.98,$$

$$\frac{\partial J}{\partial w_2} = -2((-1.4142)(50) + (0.7071)(75) + (0.7071)(90)) = -2(45.96) = -91.92.$$

Step 5 (improvement). The sensitivity ratio collapses from ≈ 576 to

$$\frac{97.98}{91.92} \approx 1.07.$$

With comparable gradient scales, a single learning rate drives both weights at similar speeds: no oscillation in w_1 while w_2 crawls – differential curvature is essentially eliminated and convergence accelerates dramatically.

Answer. (a) $(-685000, -1190)$, ratio ≈ 576 ; (b) $(-97.98, -91.92)$; (c) ratio ≈ 1.07 – balanced sensitivities $\Rightarrow$ one γ fits all, fast stable descent.

Intuition & exam tips

Housing data is the classic exam clothing for this computation. The arithmetic pattern to remember at $w = 0$: gradient $= -2 \sum y x_j$, so the *feature scale multiplies straight into the gradient*. Standardize, and the ratio drops to order 1. Quote the before/after ratio explicitly – that one number is usually a rubric line.

Problem 4 (8 marks) MODERATE COMPUTATION

Consider $L(x, y) = 9x^2 + y^2$.

- Write the GD updates for x and y as multiplications by constant factors, and find the exact range of γ for convergence. [3]
- With $\gamma = 0.1$, trace three iterations from $(x_0, y_0) = (1, 3)$. Which coordinate oscillates and why? [3]
- Compute the Hessian and its condition number. What curvature profile would make GD fastest, and which single γ then finishes in one step? [2]

Step-by-step solution

Step 1 (decoupled updates). The gradient is $\nabla L = (18x, 2y)$, so

$$x_{k+1} = (1 - 18\gamma) x_k, \quad y_{k+1} = (1 - 2\gamma) y_k.$$

Stability needs both factors inside $(-1, 1)$:

$$|1 - 18\gamma| < 1 \Rightarrow \gamma < \frac{2}{18} = \frac{1}{9}, \quad |1 - 2\gamma| < 1 \Rightarrow \gamma < 1.$$

The stiff direction binds:

$$0 < \gamma < \frac{1}{9} \approx 0.111$$

Step 2 (trace with $\gamma = 0.1$). Factors: $1 - 1.8 = -0.8$ for x ; $1 - 0.2 = 0.8$ for y :

$$(1, 3) \rightarrow (-0.8, 2.4) \rightarrow (0.64, 1.92) \rightarrow (-0.512, 1.536).$$

Both coordinates shrink by 20% per step, but x **flips sign each iteration**: its factor is negative because $\gamma = 0.1$ exceeds $1/18 \approx 0.056$, the smooth-descent threshold for that coordinate. Each step overshoots the valley floor in the stiff direction and bounces back, while y descends monotonically – zig-zag in its cleanest form.

Step 3 (Hessian and the ideal bowl).

$$H = \begin{pmatrix} 18 & 0 \\ 0 & 2 \end{pmatrix}, \quad \kappa = \frac{18}{2} = 9.$$

GD is fastest when curvature is *moderate and equal* in all directions ($\kappa = 1$, a circular bowl, e.g. $9x^2 + 9y^2$): the gradient then points straight at the minimizer, and $\gamma = 1/\lambda$ (here $1/18$) sends *both* coordinates exactly to 0 in one step. With $\kappa = 9$ no such γ exists – one-step convergence would need $\gamma = 1/18$ and $\gamma = 1/2$ simultaneously.

Answer. (a) factors $(1 - 18\gamma), (1 - 2\gamma)$; range $\gamma < 1/9$; (b) $(-0.8, 2.4) \rightarrow (0.64, 1.92) \rightarrow (-0.512, 1.536)$; x oscillates (negative factor); (c) $H = \text{diag}(18, 2)$, $\kappa = 9$; equal moderate curvature is ideal, then $\gamma = 1/\lambda$ finishes in one step.

Intuition & exam tips

Everything about differential curvature in one computation: per-coordinate factors $(1 - \gamma\lambda_i)$, the stiffest eigenvalue capping γ at $2/\lambda_{\max}$, the sign of a factor determining oscillation, and $\kappa = \lambda_{\max}/\lambda_{\min}$ as the difficulty score. When asked “why is the elliptical bowl slow?” – answer with the two factors, not with prose.

Problem 5 (6 marks) MODERATE DERIVATION + COMPUTATION

For the least-squares loss $J(\mathbf{w}) = \sum_{i=1}^n (\mathbf{w}^\top X_i - y_i)^2$:

- (a) Derive the gradient $\nabla J(\mathbf{w}) = 2 \sum_{i=1}^n (\mathbf{w}^\top X_i - y_i) X_i$. [2]
- (b) For $X_1 = (2, 1)^\top$, $y_1 = 4$ and $X_2 = (1, 3)^\top$, $y_2 = 5$, compute ∇J at $\mathbf{w}_0 = (0, 0)^\top$, take one GD step with $\gamma = 0.02$, and verify the loss decreased. [4]

Step-by-step solution

Step 1 (derive via the chain rule). Write $J = \sum_i r_i^2$ with residual $r_i = \mathbf{w}^\top X_i - y_i$. Since $\nabla_{\mathbf{w}} r_i = X_i$,

$$\nabla J = \sum_i 2 r_i \nabla_{\mathbf{w}} r_i = 2 \sum_{i=1}^n (\mathbf{w}^\top X_i - y_i) X_i.$$

Reading: each example pulls $\mathbf{w}$ along its own feature vector, in proportion to its current error.

Step 2 (gradient at $\mathbf{w}_0$). Residuals at the origin:

$$r_1 = 0 - 4 = -4, \quad r_2 = 0 - 5 = -5, \quad J_0 = 16 + 25 = 41.$$

$$\nabla J(\mathbf{w}_0) = 2[(-4)(2, 1) + (-5)(1, 3)] = 2(-8 - 5, -4 - 15) = (-26, -38).$$

Step 3 (one GD step).

$$\mathbf{w}_1 = (0, 0) - 0.02(-26, -38) = (0.52, 0.76).$$

Step 4 (verify the decrease). New residuals:

$$r_1 = 0.52(2) + 0.76(1) - 4 = -2.20, \quad r_2 = 0.52(1) + 0.76(3) - 5 = -2.20.$$

$$J_1 = (-2.2)^2 + (-2.2)^2 = 9.68 < 41. \checkmark$$

One step recovered 76% of the loss.

Answer. (a) $\nabla J = 2 \sum_i (\mathbf{w}^\top X_i - y_i) X_i$; (b) $\nabla J(\mathbf{0}) = (-26, -38)$, $\mathbf{w}_1 = (0.52, 0.76)$, $J : 41 \rightarrow 9.68$.

Intuition & exam tips

This gradient – *residual times feature vector, summed* – is the engine behind the standardization computation of Problem 3, and restricting the sum to a sample S gives the mini-batch gradient used by stochastic methods. Watch the sign convention: with $r_i = \hat{y}_i - y_i$ the formula is $+2 \sum r_i X_i$ and the minus sign lives in the update.

Problem 6 (6 marks) DIFFICULT COMPUTATION

For $F(x) = 3x^4 - 4x^3 - 12x^2 + 5$, small-step gradient descent converges to the **local** minimum at $x = -1$ from any start $x_0 < 0$, and to the **global** minimum at $x = 2$ from any start $x_0 > 0$ (the watershed – a local maximum – sits at $x = 0$). The initial point x_0 is drawn uniformly from $[-2, 3]$.

- Compute the probability that a single run converges to the global minimum. [2]
- With k independent restarts (keeping the best result), find the smallest k for which the global minimum is found with probability ≥ 0.99 . [3]
- If a separable 10-dimensional objective has such a landscape in *every* coordinate (per-coordinate success probability 0.6, independent), what is the chance that a single run lands in the global basin of *all* coordinates simultaneously? [1]

Step-by-step solution

Step 1 (one run). Success requires $x_0 > 0$ (from $x_0 < 0$ the run is funnelled to -1 ; the single point $x_0 = 0$ has probability zero under a continuous draw). Favourable length over total length:

$$p = \frac{3 - 0}{3 - (-2)} = \frac{3}{5} = 0.6.$$

Step 2 (restarts). One run fails with probability 0.4. We need

$$1 - 0.4^k \geq 0.99 \iff 0.4^k \leq 0.01.$$

Compute the powers:

$$0.4^5 = 0.01024 > 0.01, \quad 0.4^6 = 0.004096 \leq 0.01.$$

So $k = 6$ restarts (success probability ≈ 0.9959 ; $k = 5$ just misses at 0.98976).

Step 3 (ten coordinates at once). By independence:

$$P = 0.6^{10} \approx 0.006 \quad (\approx 0.6\%).$$

Even with per-coordinate odds well above one half, the joint success collapses exponentially – the high-dimensional version of the local-minima trap, and the reason restarts alone cannot rescue GD in ML.

Answer. (a) $p = 0.6$; (b) $k = 6$; (c) $0.6^{10} \approx 0.6\%$.

Intuition & exam tips

The template $1 - (1 - p)^k$ (“at least one success in k tries”) plus a careful power computation is a recurring exam device. Basins multiply across coordinates, so probabilities multiply too. Note how the numbers are engineered so that $k = 5$ *barely* fails – compute powers precisely rather than rounding early.

Problem 7 (8 marks) **DIFFICULT** **COMPUTATION**

A linear model $\hat{y} = w_1x_1 + w_2x_2$ is trained on features

$$x_1 \in \{100, 500, 900\} \text{ (e.g. sq. ft)}, \quad x_2 \in \{0.9, 0.3, 0.6\} \text{ (e.g. a rate)},$$

with $J = \sum_i (\hat{y}_i - y_i)^2$, whose Hessian is $H = 2 \sum_i X_i X_i^\top$ (rows $X_i = (x_{i1}, x_{i2})^\top$).

- Compute H and (given eigenvalues $\lambda \approx 2.14 \times 10^6$ and ≈ 1.38) the largest stable learning rate $\gamma_{\max} = 2/\lambda_{\max}$. [3]
- Min-max normalize both columns and recompute H' ; find its eigenvalues and the new $\gamma'_{\max}$. [4]
- By what factor did the admissible learning rate improve? [1]

Step-by-step solution

Step 1 (raw Hessian). The needed sums:

$$\begin{aligned} \sum x_{i1}^2 &= (1 + 25 + 81) \times 10^4 = 107 \times 10^4, & \sum x_{i1}x_{i2} &= 90 + 150 + 540 = 780, \\ \sum x_{i2}^2 &= 0.81 + 0.09 + 0.36 = 1.26. \end{aligned}$$

Hence

$$H = 2 \begin{pmatrix} 1070000 & 780 \\ 780 & 1.26 \end{pmatrix} = \begin{pmatrix} 2.14 \times 10^6 & 1560 \\ 1560 & 2.52 \end{pmatrix}.$$

With the given $\lambda_{\max} \approx 2.14 \times 10^6$ ($\approx H_{11}$; the off-diagonal is tiny in comparison) and $\lambda_{\min} \approx 1.38$:

$$\gamma_{\max} = \frac{2}{\lambda_{\max}} \approx \frac{2}{2.14 \times 10^6} \approx 9.3 \times 10^{-7}, \quad \kappa \approx \frac{2.14 \times 10^6}{1.38} \approx 1.6 \times 10^6.$$

Any learning rate above $\sim 10^{-6}$ blows up – and at $\sim 10^{-6}$ the w_2 direction (curvature ≈ 1.4) moves glacially.

Step 2 (min-max scale the columns). Column 1: min = 100, range = 800:

$$x'_1 = (0, 0.5, 1).$$

Column 2: min = 0.3, range = 0.6:

$$x'_2 = (1, 0, 0.5).$$

Step 3 (new Hessian). New sums:

$$\sum x_1'^2 = 0 + 0.25 + 1 = 1.25, \quad \sum x_1'x_2' = 0 + 0 + 0.5 = 0.5, \quad \sum x_2'^2 = 1 + 0 + 0.25 = 1.25.$$

$$H' = 2 \begin{pmatrix} 1.25 & 0.5 \\ 0.5 & 1.25 \end{pmatrix} = \begin{pmatrix} 2.5 & 1 \\ 1 & 2.5 \end{pmatrix}.$$

Eigenvalues of a symmetric $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ matrix are $a \pm b$:

$$\lambda = 2.5 \pm 1 = \{3.5, 1.5\}, \quad \gamma'_{\max} = \frac{2}{3.5} \approx 0.571, \quad \kappa' = \frac{3.5}{1.5} \approx 2.33 \quad (\text{was } \sim 10^6).$$

Step 4 (improvement factor).

$$\frac{\gamma'_{\max}}{\gamma_{\max}} \approx \frac{0.571}{9.3 \times 10^{-7}} \approx \boxed{6 \times 10^5}.$$

Scaling the features let the learning rate grow by roughly six orders of magnitude while *also* balancing the two directions – the quantitative content of “normalize your features before gradient descent”.

Answer. (a) $H = \begin{pmatrix} 2.14 \times 10^6 & 1560 \\ 1560 & 2.52 \end{pmatrix}$, $\gamma_{\max} \approx 9.3 \times 10^{-7}$; (b) $H' = \begin{pmatrix} 2.5 & 1 \\ 1 & 2.5 \end{pmatrix}$, $\lambda = \{3.5, 1.5\}$, $\gamma'_{\max} \approx 0.571$; (c) $\approx 6 \times 10^5 \times$ larger.

Intuition & exam tips

The grand synthesis: feature scales enter the Hessian *squared* (x^2 terms), so a 10^3 scale gap becomes a 10^6 curvature gap, which caps γ at $2/\lambda_{\max} \sim 10^{-6}$. Normalization is not cosmetic – it is a 10^5 – $10^6 \times$ speedup you can compute. Handy fact used in (b): eigenvalues of $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ are $a \pm b$ – worth memorizing for 2×2 symmetric matrices with equal diagonal.

Highest-yield patterns: Problems 3 (standardization vs gradients), 4 (curvature factors) and 7 (Hessian before/after scaling) – most exam variants are re-skins of these.